

Casting or Fabrications of Aircraft Engine Technology

ME 3633 Manufacturing Processes

Project #11

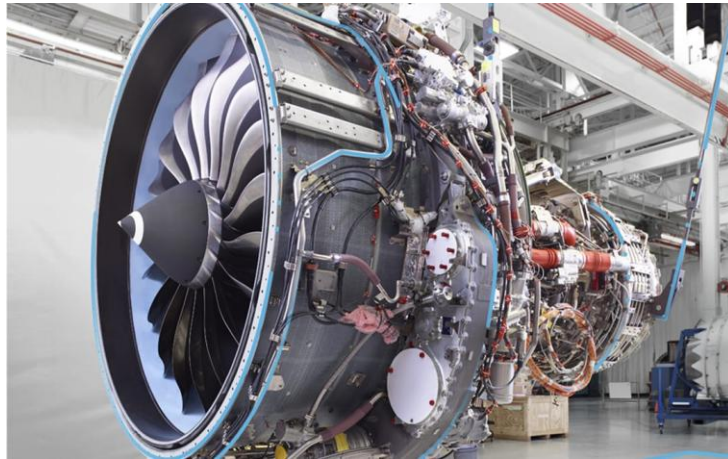


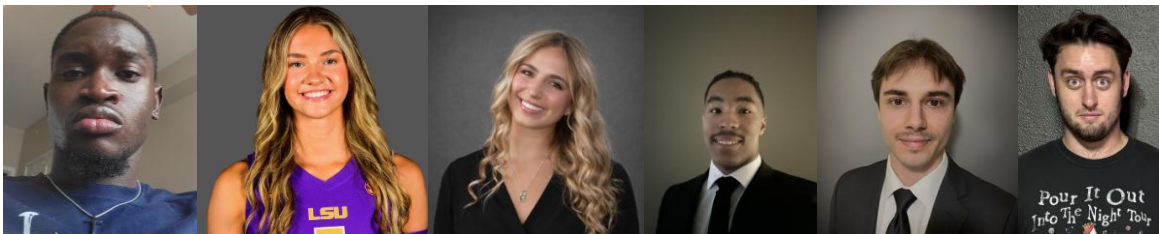
Figure 1. Representative aircraft engine used as a visual reference for the project.

Student Names: Neal Butler, Jessica Jones, Marie Rohli, Anthony Graham, Jean-Paul Northcutt,
Parker Gilfour

Group #: 11

Instructor: Muhammad Wahab

Date: 3/20/2026



Abstract

Aircraft engines operate in extreme environments characterized by high temperatures, pressures, oxidation, and cyclic loading, requiring a carefully controlled combination of advanced manufacturing processes to ensure component performance and safety. The selection of manufacturing methods is not arbitrary, but instead is driven by the specific thermal, mechanical, and fatigue requirements of each engine region. This report examines the primary manufacturing routes used for major hot section and structural components in modern turbofan engines, with emphasis on why specific processes are selected for particular applications.

Vacuum investment casting is analyzed for its critical role in producing complex, single crystal nickel superalloy turbine blades and vanes, where internal cooling passages and superior creep resistance are essential. In contrast, fabrication methods such as premium forging, precision machining, and advanced joining are evaluated for their ability to produce high fatigue, safety critical components including titanium rotor disks and compressor assemblies. The growing role of additive manufacturing is also discussed, particularly in enabling part consolidation and complex internal geometries that cannot be achieved through conventional processes.

In addition to process selection, this report highlights the inherent trade offs associated with each manufacturing route, including cost, defect sensitivity, and process control challenges. Common defects such as shrinkage porosity, stray grains, residual stress distortion, and recast layers are examined in detail, along with their impact on component life and reliability. Finally, the report emphasizes the critical role of inspection, process validation, and traceability through techniques such as Computed Tomography (CT), Fluorescent Penetrant Inspection (FPI), and rigorous documentation practices. These measures are essential to detect manufacturing induced anomalies and ensure long term flight safety in aerospace applications.

Contents

Abstract	2
List of Figures	3
2. Where casting and fabrication are used in an aircraft engine	4
3. Casting technology for turbine blades and vanes	5
4. Blade architecture: why casting alone is not enough	6
5. Fabrication methods used for other engine parts	7
6. Additive manufacturing as a complement.....	7
7. Main advantages of advanced casting and fabrication routes	8
8. Main disadvantages and trade-offs.....	9
9. Defects that can arise in cast parts.....	10
10. Defects that can arise in fabricated parts.....	11
11. How manufacturers reduce or solve casting defects	11
12. How manufacturers reduce or solve fabrication defects.....	11
13. Inspection, traceability, and process validation	12
14. Repair and life extension	13
15. Conclusion	13
References.....	13

List of Figures

Figure 1. Representative aircraft engine used as a visual reference for the project.	1
Figure 2. Typical match between engine region, material, and manufacturing route.	4
Figure 3. Simplified concept of directionally solidified / single-crystal blade casting.	5
Figure 4. Example of internal cooling path geometry relevant to turbine blade manufacture.	6
Figure 5. Example of a fuel nozzle component used to illustrate additive manufacturing in aerospace.	8
Figure 6. CT style porosity image used to illustrate internal casting defects.	10
Figure 7. Common casting defects relevant to hot section hardware.	11

1. Introduction

Aircraft engine technology is one of the most demanding areas in manufacturing because the final parts must survive high temperature, pressure, vibration, oxidation, and cyclic loading while still maintaining very tight dimensional control. A modern engine therefore cannot rely on a single process. Instead, it combines casting, forging, sheet metal fabrication, machining, drilling, joining, coatings, and non destructive inspection into one controlled manufacturing system.

The prompt asks for a discussion of casting or fabrication methods used in aircraft engines, the advantages and disadvantages of those methods, the defects that may arise, and the ways manufacturers reduce or solve those problems. This report focuses on the major hot section and structural parts of a turbofan engine and explains where casting dominates, where fabrication dominates, and why strong inspection and process control are essential.

2. Where casting and fabrication are used in an aircraft engine

A turbofan engine contains several major modules: the fan, compressor, combustor, turbine, and outer structural cases. Different modules use different materials and manufacturing routes because the service conditions are not the same across the engine.

Engine area	Typical material	Manufacturing route used most often
Fan / Front compressor	Titanium alloys	Forging, machining, composite forming for some fan systems
Rear compressor	Titanium, steel, nickel alloys	Forged disks, machined blisks, fabricated casings
Combustor	Nickel based sheet alloys	Sheet forming, welding, brazing, drilling, coatings
Turbine blades / vanes	Nickel superalloys	Vacuum investment casting, heat treatment, coating, finish machining
Disks / shafts	Titanium or nickel alloys	Premium forging, machining, shot peening, inspection

Figure 2. Typical match between engine region, material, and manufacturing route.

From a manufacturing perspective, the turbine is the area where casting is most important because airfoils must withstand the highest temperatures. In contrast, disks, shafts, many compressor parts, and many combustor structures are fabricated or machined from

wrought material because they require predictable fatigue performance and high dimensional stability.

3. Casting technology for turbine blades and vanes

The most important casting process in aircraft engines is vacuum investment casting of turbine blades and vanes. This process is used because these parts are geometrically complex and must include hollow cooling passages that would be extremely difficult or impossible to produce by conventional machining from solid stock.

A simplified process chain is shown below. First, a wax pattern is created around a ceramic core. The core creates the internal cooling passage geometry. The pattern is coated repeatedly to build a ceramic shell. After dewaxing, the mold is fired, filled with molten superalloy under vacuum, and then solidified in a controlled way. Depending on the component, the structure may be equiaxed, directionally solidified, or single crystal. The shell and core are removed, the part is heat treated, coated, drilled or machined where required, and finally inspected.

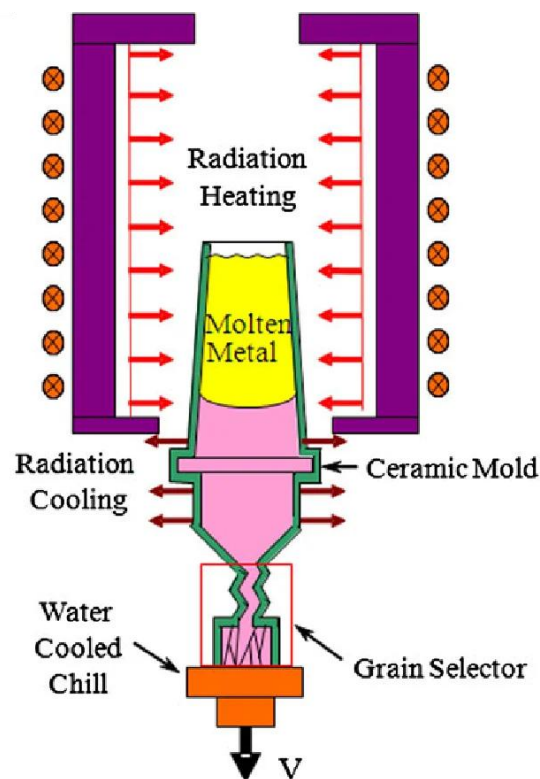


Figure 3. Simplified concept of directionally solidified / single-crystal blade casting.

Single crystal and directionally solidified blades are especially important in the hottest turbine stages because grain boundaries are weak points for creep and thermal fatigue. Removing or controlling those grain boundaries improves high temperature capability.

However, it also makes the process more difficult, more expensive, and more sensitive to solidification defects.

The selection of investment casting for turbine blades is primarily driven by the need to balance geometric complexity with high temperature performance. Conventional machining cannot produce the intricate internal cooling passages required for modern airfoils, and alternative fabrication methods would introduce joints or interfaces that reduce reliability under extreme thermal loading. Casting therefore provides not only a manufacturing solution, but also a performance advantage by enabling integrated geometries that improve cooling efficiency and extend component life.

4. Blade architecture: why casting alone is not enough

Even when the base airfoil shape is cast, the blade is still not finished. The finished part must include a precisely positioned root and platform, internal cooling passages, film cooling holes, and a surface coating system. The cooling holes are commonly added later by precision laser drilling or EDM. Thermal barrier coatings are applied after heat treatment to reduce the metal temperature at the surface. Final root surfaces and sealing features are also machined after casting.

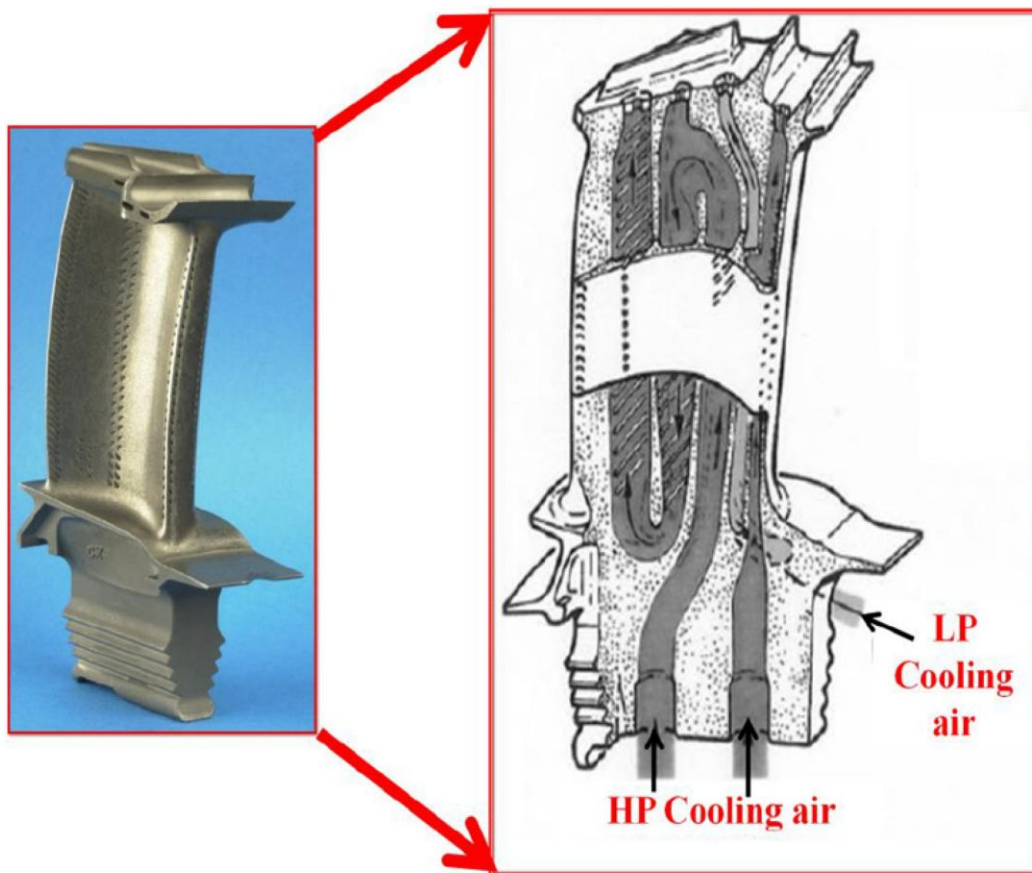


Figure 4. Example of internal cooling path geometry relevant to turbine blade manufacture.

This means a turbine blade should be viewed as a complete manufacturing system rather than only a cast part. Casting, drilling, coating, heat treatment, dimensional control, and inspection all have to be matched to one another. A weakness in any step can reduce the life of the entire component.

5. Fabrication methods used for other engine parts

Many aircraft engine parts are better described as fabricated or machined rather than cast. Rotor disks are generally forged from premium titanium or nickel alloys because the wrought microstructure gives strong fatigue performance. Compressor blisks may be machined from solid or produced by joining blades to a disk. Combustor liners and cases are often fabricated from sheet or plate by forming, welding, brazing, and drilling. Casings and structural rings may be formed, rolled, machined, and welded.

Fabrication routes are favored when the part requires excellent fatigue properties, thin walled sheet geometry, repairability, or a microstructure that is easier to qualify than an equivalent casting. The trade off is that fabricated parts may require more machining, more joining steps, and more process control to avoid distortion or local defects.

Fabrication processes are selected over casting when fatigue performance, structural integrity, and material consistency are more critical than geometric complexity. Wrought materials produced through forging provide a refined and directional grain structure that improves resistance to crack initiation and propagation. This is especially important for rotating components such as disks and shafts, where failure would have catastrophic consequences. As a result, fabrication is not simply an alternative to casting, but a necessary approach for components where reliability and fatigue life dominate design requirements.

6. Additive manufacturing as a complement

Additive manufacturing is not a replacement for all casting or fabrication processes, but it is becoming a valuable complement. A well known example is the LEAP fuel nozzle, where GE Aerospace reported that additive manufacturing consolidated about twenty previously joined pieces into one printed part. The company also reported a substantial weight reduction and improved durability. These are exactly the kinds of gains that make additive manufacturing attractive in aircraft engines: part consolidation, improved internal geometry, and lower assembly complexity.



Figure 5. Example of a fuel nozzle component used to illustrate additive manufacturing in aerospace.

Even so, additive manufacturing still requires tight control of powder quality, build parameters, heat treatment, post processing, and inspection. For many large structural or hot section parts, casting and fabrication remain the dominant methods. Additive manufacturing is most useful when the geometry complexity or assembly reduction clearly offsets the process cost and qualification burden.

7. Main advantages of advanced casting and fabrication routes

Near net shape casting makes extremely complex hot section geometry possible. An example of such geometry is hollow turbine airfoils with internal cooling passages, a crucial structure for regulating thermal load and therefore increasing durability. By forming this structure during casting solidification, the need for post processing is reduced and there is less material waste.

Single crystal and directionally solidified casting are methods used to improve durability and offer high temperature capability. Single crystal casting of metal components uses a single continuous crystal lattice to reduce grain boundaries. Directional solidification is a process where the metal cast is solidified in a single direction, often from the bottom to the top. This aligns the grain boundaries in the direction of primary stress, ensuring that crack initiation will not occur transverse to applied stress. Both methods significantly increase creep resistance and thermal fatigue for components that will endure high temperatures.

Forged rotor parts provide excellent fatigue performance for safety critical disks and shafts. During the forging process, grains are elongated and aligned through the plastic deformation of the metal. For rotors, this grain flow follows a circular direction which aligns with the applied stress. This drastically reduces crack propagation and offers a longer lifespan of the part.

Fabrication methods such as forming, welding, and brazing are effective for combustors, cases, and repairable structures. This is because these parts are all made of very thin metal which makes casting methods difficult and prone to defects. The fabrication of these parts allows for precise wall thickness and better dimensional control. The use of welding enables modular construction, where smaller, more optimized parts can be joined together into a full assembly. This also allows for damaged parts to be easily replaced or repaired. Forming and

brazing cause better thermal gradients than casting, reducing the risk of internal stresses that can cause shrinkage or failure.

Additive manufacturing can reduce part count, save weight, and simplify assemblies when used in the right location. Complex geometry, such as internal passages and supports, can all be created simultaneously instead of separately. Some internal features are impossible to create via casting or machining. Creating structures as one piece eliminates the need for mechanical fasteners and joints, reducing potential failure and stress concentration points.

Repair processes can extend life for expensive engine hardware instead of replacing the part immediately. The removal of damaged parts suffering from thermal fatigue, oxidation, or general wear prevents further propagation. Processes such as blending and crack removal eliminate small defects before they become larger, thereby extending the life of the part. Some components can be simply restored through brazing and welding methods which can rebuild worn areas and return original dimensions. Maintenance such as reapplying thermal barrier and oxidation-resistant coatings prevent further damage and lets the part survive more extreme conditions.

Aircraft engines can quickly take on extreme geometries best produced and reproduced by casting and additive manufacturing. Thin hollow bodies, such as those found in various air flow pipes in a jet engine, can be difficult to produce from techniques like milling or turning, but by designing molds for casting these parts, manufacturers can reliably achieve desirable geometries. Similarly, additive manufacturing methods can create advanced geometries, even some which were once impossible as homogeneous parts (particularly useful for liquid fuel rocket engines). This ability to produce single parts that previously would have been multiple parts with interfaces significantly reduces complexities, points of failure, and energy losses.

Additionally, processes such as welding parts can improve the ability to repair damaged parts. For key components of the engine, it may be best to design a part to ideally have an infinite life span, but that trends towards higher costs and reliability. Meaning, where applicable, it can at times be more effective to design parts to have maintenance that uses techniques like welding to maintain its operation.

8. Main disadvantages and trade-offs

- Investment casting is expensive and highly sensitive to shell quality, melt cleanliness, thermal conditions, and core position.
- Single crystal casting has very tight process windows and can be limited by stray-grain formation, freckles, or orientation issues.
- Fabrication routes can introduce distortion, residual stress, weld defects, recast layers, or local damage around drilled features.
- Additive manufacturing still faces challenges with surface finish, defect characterization, and certification effort.

- All advanced aerospace routes require strong inspection and traceability, which increases cost and documentation burden.

One of the most significant trade offs associated with advanced manufacturing processes is the increased reliance on inspection and validation. As process complexity increases, so does the risk of subtle defects that may not be visible without advanced non destructive evaluation techniques. This introduces additional cost, time, and documentation requirements, as every critical component must be verified through multiple inspection stages. In this sense, the advantages of high performance manufacturing processes are inseparable from the burden of ensuring their reliability through rigorous quality control systems.

9. Defects that can arise in cast parts

The most common casting problems in aircraft engine parts are porosity, shrinkage, inclusions, dross, hot tears, core shift, stray grains, freckles, and dimensional error. These defects matter because they can reduce fatigue life, reduce creep resistance, change cooling effectiveness, or cause local overheating in service.

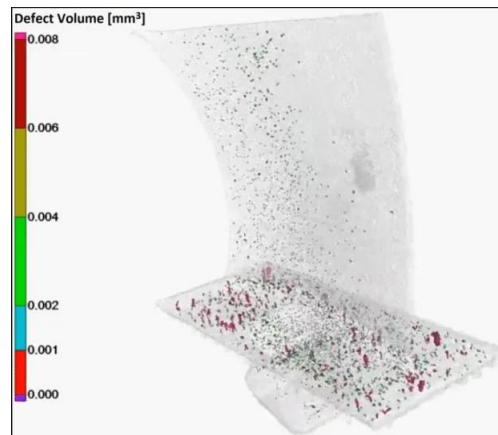


Figure 6. CT style porosity image used to illustrate internal casting defects.

Defect	Typical effect on part	Possible cause
Porosity / shrinkage	Stress concentration and lower fatigue life	Gas entrapment, poor feeding, local hot spot
Inclusions / dross	Crack initiator or poor surface quality	Melt contamination, shell debris, oxidation
Hot tear	Crack during cooling or solidification	Constrained contraction and thermal stress

Stray grains / freckles	Reduced single crystal quality	Incorrect thermal gradient or chemistry segregation
Core shift	Wrong wall thickness or cooling mismatch	Poor core support or wax variation

Figure 7. Common casting defects relevant to hot section hardware.

10. Defects that can arise in fabricated parts

Fabricated and machined parts experience a different set of problems. Drilled holes can develop recast layers, burrs, white layer, or geometrical drift. Welded or brazed parts can show distortion, lack of fusion, brittle zones, contamination, or rework damage. Blisks and repaired structures may also suffer from alignment error, local overheating, or unacceptable blending that changes the stress field of the part.

These issues are especially important in aircraft engines because fabricated components are often highly stressed or are expected to guide cooling flow very precisely. A small process error can therefore affect fatigue performance, hot section life, or dimensional fit during assembly.

11. How manufacturers reduce or solve casting defects

- Use strong upfront design rules for wall thickness, fillets, feeding, core support, and grain selection geometry.
- Control wax quality, shell materials, furnace conditions, vacuum level, pouring practice, and withdrawal rate.
- Maintain melt cleanliness and minimize contamination from tools, shells, crucibles, or the environment.
- Apply heat treatment and, where qualified, HIP to improve internal soundness after casting.
- Inspect the part with dimensional checks, fluorescent penetrant inspection, X-ray, CT, metallography, and acceptance criteria appropriate for flight hardware.

The key idea is that casting defects are not solved by inspection alone. Good results come from process design first and inspection second. Inspection confirms that the process worked; it does not replace good process control.

12. How manufacturers reduce or solve fabrication defects

- Lock down process windows for machining, drilling, welding, brazing, and heat treatment.
- Control tool wear, pulse energy, coolant, fixturing, shielding gas, and alignment.

- Use qualified procedures for welded or brazed repair and restore geometry only within approved limits.
- Reestablish coolinghole quality and coatings after repair when the procedure requires it.
- Verify fabricated parts with dimensional inspection, borescopes, FPI, eddy current, radiography, or CT as appropriate.

• Lock down process windows have been implemented within the aerospace manufacturing sector to aggressively minimize fabrication defects. Lock down process windows are strict ranges of physical properties to stay within during a fabrication process. For example, when machining a part, the temperature of the part must not exceed a given temperature to not induce thermal fatigue within the part.

These limits extend beyond properties of the desired part, but also to the tools producing them. Minimized wear of tool parts, precision of operations, and general best practices are all expected to be rigorously upheld. This could look like using specific welding technologies over another or highly tracked usage of milling bits.

Often, it is not enough to hope these fabrication practices worked exactly as they ought to. This means parts are often inspected using various techniques to catch manufacturing failures before they go into operation. This can be as simple as insuring geometric measurements were successfully obtained, or as complicated as Fluorescent Penetrant Inspection (FPI) which involves multiple steps to make surface defects highly.

13. Inspection, traceability, and process validation

In aircraft engine manufacturing, evidence is part of the product. A part is not acceptable simply because it looks correct; it must also be supported by records showing material lot, process route, heat treatment history, operator control, machine parameters, and inspection results. This is especially important for critical rotating parts such as disks and shafts because manufacturing induced anomalies can become crack starters that are not acceptable in service.

For this reason, aerospace manufacturers depend heavily on process validation, qualified procedures, and non-destructive evaluation. The FAA and industry guidance on critical rotating parts make it clear that manufacturing anomalies must be addressed through both robust process control and robust detection capability.

In addition to defect detection, inspection data is increasingly used to improve manufacturing processes themselves. Feedback from CT scans, metallographic analysis, and dimensional inspection can be used to refine process parameters, reduce variability, and prevent recurring defects. This creates a closed loop manufacturing system in which inspection is not only a verification tool but also a driver for continuous improvement. As

aerospace manufacturing moves toward digital integration, inspection data is becoming a critical part of process optimization and lifecycle management.

14. Repair and life extension

Aircraft engine parts are expensive, so repair strategy is often built into the manufacturing strategy. Depending on the part and the approved procedure, manufacturers may blend small damage, braze certain cracks, restore geometry, redrill cooling holes, recoat surfaces, or replace only selected segments instead of scrapping the entire assembly. However, repair is never automatic. It must remain within engineering limits and must be followed by inspection that proves the repaired hardware is acceptable.

15. Conclusion

Aircraft engine technology depends on both casting and fabrication, not one or the other. Vacuum investment casting is the key route for turbine blades and vanes because it can create hollow superalloy airfoils with internal cooling passages. Fabrication and machining dominate many other engine parts, especially disks, shafts, blisks, combustor hardware, and structural cases. Additive manufacturing now supports these routes where geometry complexity and part consolidation justify it.

The biggest lesson from this project is that advanced engine manufacture is controlled by the relationship between material, process history, defects, inspection, and service environment. The best process is not simply the cheapest or the newest one. An additional key takeaway is that manufacturing and inspection cannot be treated as separate activities. Instead, they must function as an integrated system in which process selection, defect control, and verification methods are all aligned with the intended service environment of the component. It is the process that can repeatedly produce the required microstructure, geometry, and evidence of quality for a safety critical aircraft engine component.

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